

AKTIO, Greece

WMO: 166430

Lat: 38.925N

Lon: 20.765E

Elev: 11

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 37.2	(c) 39.5	(d) 19.4	(e) 14.6	(f) 42.8	(g) 23.7	(h) 17.9	(i) 44.0	(j) 30.5	(k) 49.9	(l) 26.2	(m) 50.6	(n) 9.2	(o) 40	(p) 0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 16.4	(c) 91.5	(d) 72.4	(e) 89.2	(f) 73.2	(g) 86.6	(h) 73.5	(i) 77.6	(j) 85.4	(k) 76.4	(l) 84.2	(m) 75.4	(n) 82.9	(o) 9.4	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	132.8	81.9	73.8	126.0	80.9	73.2	123.2	80.5	41.0	85.3	39.9	84.6	38.9	82.9	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.5	(b) 20.6	(c) 18.1	(e) 33.2	(f) 97.9	(g) 2.4	(h) 2.7	(i) 31.5	(j) 99.8	(k) 30.0	(l) 101.4	(m) 28.7	(n) 102.9	(o) 26.9	(p) 104.9		
			DB													
			WB													
				29.1	80.2	2.9	1.6	27.0	81.3	25.4	82.3	23.7	83.2	21.6	84.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.8	50.0	51.1	54.4	59.4	66.7	73.9	78.0	79.3	73.3	66.8	59.3	52.5
	DBStd	11.11	4.63	4.73	4.19	4.32	4.01	4.49	3.31	3.58	3.93	4.12	4.98	4.78
	HDD50	140	55	40	11	1	0	0	0	0	0	0	4	29
	HDD65	1988	465	390	330	176	28	1	0	0	1	29	179	389
	CDD50	5173	55	70	147	282	518	717	868	907	700	521	282	106
	CDD65	1541	0	0	0	6	80	268	403	442	250	85	7	0
	CDH74	12036	0	0	1	28	311	1963	3715	4268	1525	216	9	0
	CDH80	3700	0	0	0	3	36	537	1234	1558	321	11	0	0
Wind	WSAvg	8.0	8.3	8.8	8.4	7.8	7.6	7.8	8.1	7.8	7.5	7.4	7.7	8.3
Precipitation	PrecAvg	35.4	4.3	4.1	3.3	2.3	1.4	0.5	0.3	0.5	2.7	4.0	5.7	6.2
	PrecMax	51.4	11.2	8.5	6.6	6.5	4.0	1.9	1.9	2.8	6.9	12.2	13.3	11.8
	PrecMin	21.1	0.0	0.4	0.0	0.5	0.2	0.0	0.0	0.0	0.2	0.0	0.0	0.8
	PrecStd	7.5	2.8	2.3	1.7	1.5	0.9	0.4	0.4	0.7	1.9	2.6	2.6	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	67.7	70.0	77.3	84.2	93.2	93.7	96.6	89.3	81.2	75.1	67.9
		MCWB	56.6	55.6	57.3	62.5	68.2	71.1	71.4	69.8	70.6	67.5	64.8	60.4
	2%	DB	61.2	62.9	66.2	73.1	80.2	88.2	90.7	92.7	85.9	78.5	71.6	64.4
		MCWB	55.1	55.3	57.1	60.7	66.7	72.8	73.0	71.8	72.4	67.1	63.3	58.7
	5%	DB	59.1	60.8	64.0	69.8	77.2	85.7	88.1	89.6	83.8	76.6	69.4	62.3
		MCWB	53.6	54.5	56.6	60.0	66.7	72.8	73.8	73.2	71.5	66.5	62.0	57.0
	10%	DB	57.2	58.9	62.1	67.6	75.1	82.7	86.1	87.5	81.0	74.4	66.6	59.5
		MCWB	52.1	53.0	55.8	59.5	66.1	71.7	73.8	73.7	70.3	65.6	60.2	54.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.6	58.0	60.3	65.7	71.9	77.9	78.9	79.7	76.7	71.8	67.1	62.1
		MCDB	62.5	63.0	65.2	72.9	79.0	85.6	87.1	86.8	84.0	77.3	72.7	65.6
	2%	WB	56.2	56.4	58.8	63.2	69.7	75.6	77.4	78.0	74.9	69.9	64.5	59.3
		MCDB	60.3	61.1	63.8	69.5	76.4	83.7	85.4	85.5	81.5	75.2	69.9	63.1
	5%	WB	54.4	55.1	57.5	61.5	68.3	74.3	76.3	76.7	73.6	68.6	62.8	57.6
		MCDB	58.3	59.8	62.5	67.3	74.7	82.1	84.1	84.4	80.3	73.8	67.6	61.3
	10%	WB	52.7	53.7	56.2	60.1	66.9	72.8	75.2	75.6	72.3	67.1	61.2	55.7
		MCDB	56.3	58.3	60.8	65.6	73.2	80.4	82.7	83.1	78.9	72.3	65.7	59.1
Mean Daily Temperature Range	5% DB	MDBR	11.4	12.3	13.3	14.3	15.0	15.8	16.2	16.4	14.9	13.6	12.3	11.1
		MCDBR	12.8	14.2	15.4	17.8	18.2	19.5	19.4	19.9	17.3	15.5	14.0	12.5
		MCWBR	8.5	8.8	9.2	9.5	9.2	8.5	8.8	9.2	8.2	8.0	7.8	8.4
	5% WB	MCDBR	11.6	13.3	14.0	15.8	16.5	17.5	17.3	17.4	15.7	14.1	12.8	11.0
		MCWBR	8.5	8.9	9.0	9.5	9.5	8.6	8.7	8.8	8.3	8.3	8.0	8.4
Clear-Sky Solar Irradiance	taub		0.352	0.382	0.427	0.474	0.464	0.448	0.453	0.467	0.442	0.414	0.381	0.359
	taud		2.436	2.324	2.189	2.073	2.161	2.256	2.248	2.202	2.265	2.315	2.373	2.408
	Ebn at Noon		258	264	262	256	261	265	262	255	254	249	243	245
	Edh at Noon		27	34	43	51	47	43	43	44	39	34	28	26
All-Sky Solar Radiation	RadAvg		624	878	1297	1710	2077	2362	2428	2139	1589	1112	736	563
	RadStd		69	102	117	139	107	115	87	79	95	87	63	84

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.67	N/A	N/A	+2.20	N/A	N/A	N/A	-166	+263	+151
(47) Regional (8 neighbors)		+0.58	N/A	N/A	+1.57	N/A	N/A	N/A	-160	+209	+108

Nomenclature: See separate page